

CODE APEX

HACK THE UPSIDE DOWN

Team Details

- **Team name – DietCode**
- **Team leader Name – Harsh Gidwani**
- **Problem Statement (Domain) – 1.3 , Automated Social Media Content Generation.**

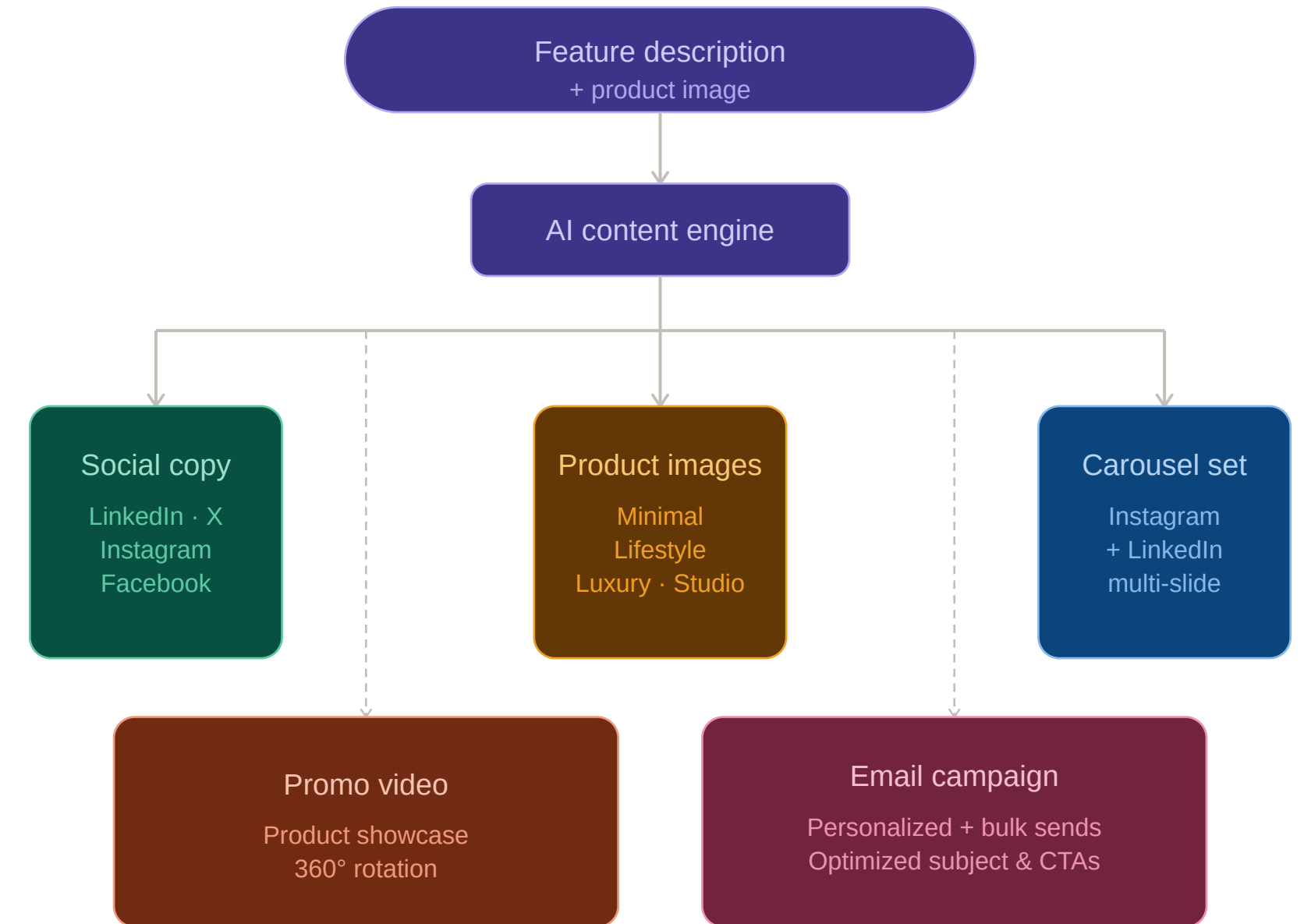
Brief about the idea

FEATURE-TO-FEED, ZERO-LATENCY MARKETING APPROACH

The gap: Every time a company ships a new feature, the marketing clock starts ticking, but the content pipeline doesn't. (it never will without our tool).

**WHAT WE'RE BUILDING: ONE INPUT. EVERY PLATFORM.
EVERY FORMAT. UNDER 2 MINUTES.**

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differentiator : Every publish makes the next generation smarter — post-publish engagement data (views, likes, comments) flows back into the RAG memory, so the system learns what actually works for that brand's audience. Built for product teams like Commudle who ship features repeatedly, not once.



One input. Every format. Every platform.

All outputs generated under 2 minutes

Proposed Solution Framework

STAGE 1 – CONTEXT INGESTION

Before generating anything, the system retrieves brand context.

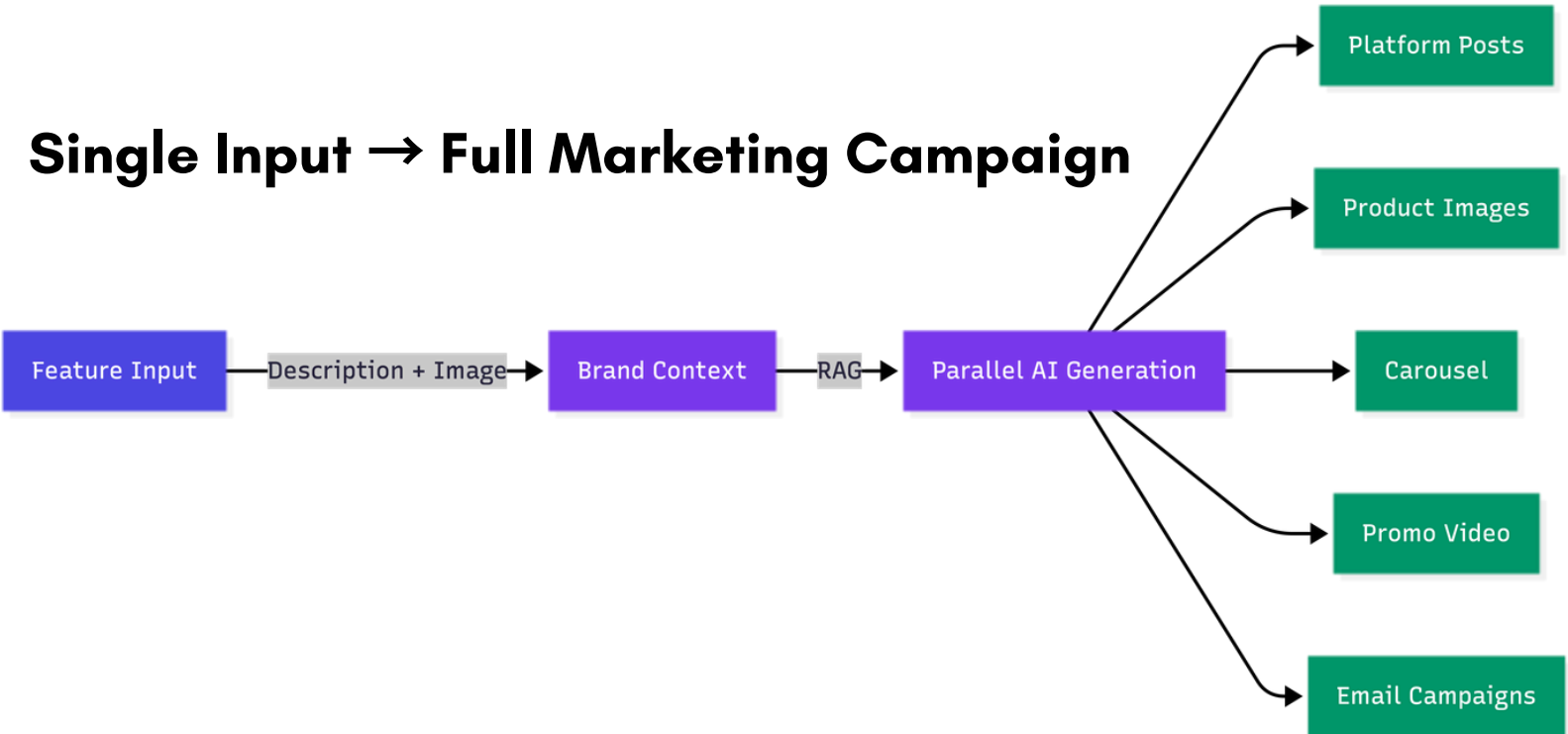
Company documents, brand guidelines, tone preferences, audience definitions, and past content are embedded in a vector database. On every generation request, the most relevant brand context is pulled via similarity search and injected into every prompt.

STAGE 2 – PARALLEL MULTI-MODAL GENERATION

Multiple engines fire simultaneously, not sequentially:

Copy Engine	4 platform-specific posts: LinkedIn (professional), X (punchy), Instagram (emoji-rich, CTA-driven), Facebook
Visual Engine	4 product image styles (Minimal white-bg, Lifestyle in-context, Luxury dramatic lighting, Studio professional) + carousel image set
Email engine	AI-generated personalized email campaigns at scale with optimized subject lines and CTAs.
Video Engine	Product showcase video + rotation clip from a single input image

Single Input → Full Marketing Campaign



STAGE 3 – CONTENT PACKAGING

All generated assets are bundled into a single preview dashboard. Each piece is editable, tweak the LinkedIn copy, swap an image style, regenerate the video prompt. Once approved, export as a ready-to-publish package or push directly to connected social accounts.

WHY DO WE WIN?

Why not just use ChatGPT + Buffer? Because neither knows your brand, your past content, or what performed. Every output here is RAG-grounded in your own docs and gets sharper with every publish — that's not a tool, that's a system that compounds.

Technical Approach & System Architecture

RAG MEMORY

- Documents, brand guidelines, past content → **chunked** → **embedded (Gemini Embeddings)** → stored in Qdrant vector database
- **Every generation call** → **top-K context retrieval** → **injected into prompt**

MULTI MODAL GEN

- **Text: Gemini 2.0 Flash**
generates platform, adapted copy with tone, length, and format.
- **Images: Gemini Image**
Generation, creates product photos across 4 visual styles from a single product image
- **Video: Replicate** (image-to-video + MiniMax Director 360°) generates short-form video from one static image
- **AI-generated personalized email campaigns** with dynamic subject lines, CTAs, and bulk lead targeting.

JOB PIPELINE

- 3 — **Async Job Pipeline**
 - BullMQ + Redis manages generation jobs
 - **Real-time status polling** — UI shows progress per asset type
 - **Graceful degradation** — if video fails, the rest of the package still delivers

SOCIAL PLATFORM

- OAuth2 integrations with **LinkedIn API v2**, **Instagram Graph API**, **YouTube Data API v3**
- **Post-publish engagement metrics** (views, likes, comments) pulled back into analytics dashboard

Brand Input (Docs, Guidelines, Past Content)

RAG Memory (Gemini Embeddings + Qdrant DB)

Multi-Modal Gen (Text · Image · Video via Gemini + Replicate)

Job Pipeline (BullMQ + Redis Async Queue)

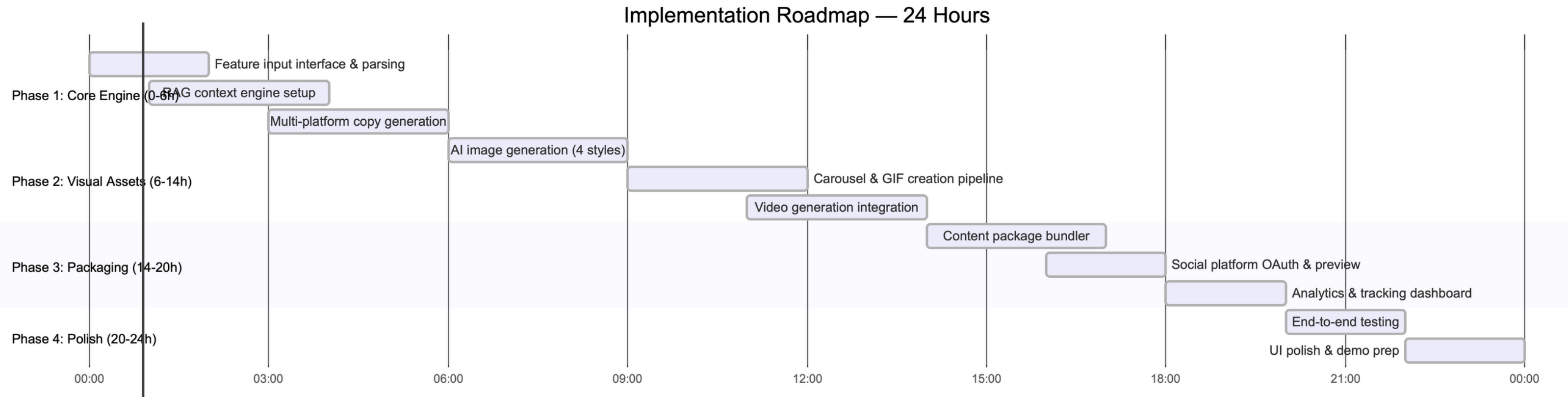
Social Publish (LinkedIn · Instagram · YouTube)

Analytics (Engagement Metrics Dashboard)

Technology Stack of the Solution

layer	Technology	Details
AI GENERATION	Gemini 2.0 Flash, Replicate (Wan 2.1 + MiniMax Director)	Single model for both text & image gen. 3 video pipelines — image-to-video, 360° rotation, multi-angle FFmpeg composition. Parallel execution via async.
BRAND MEMORY (RAG)	LangChain, Qdrant, Gemini Embeddings (768-dim)	Docs/PDFs chunked → embedded → vector stored. Top-K similarity retrieval on every generation call. Sub-100ms lookup. Ensures brand-consistent output
SOCIAL PLATFORM LAYER	LinkedIn API v2, Instagram Graph API, YouTube Data API v3	Real OAuth2 — not mocked. Engagement analytics (views, likes, comments) pulled post-publish. Token refresh + data caching built in
FRONTEND	Next.js 15, React 19, Tailwind v4, shadcn/ui, Framer Motion, Recharts	Server components + streaming for real-time generation progress. Analytics dashboards with interactive charts. Responsive across devices. Live output preview dashboard shows a real generated content package before export.
INFRASTRUCTURE	Prisma , BullMQ + Redis, Cloudinary, Upstash.	Async job pipeline for parallel generation. Cloudinary handles all media CDN + transforms. Rate limiting via sliding window. One-command deployment

Implementation Roadmap



0-8 hours	Feature input form with description + image upload. RAG context engine (Qdrant + embeddings). Multi-platform copy generation with tone/format controls. AI product image generation (4 styles via Gemini).
Hours 8-18	Video generation integration (Replicate Wan 2.1 + MiniMax Director). Carousel image creation pipeline. Content package bundler — preview all assets in one view. Social platform OAuth connections. Export/download as ready-to-publish bundle.
Hours 18-24	End-to-end pipeline testing across all generation engines. Edge case handling and graceful degradation. UI polish. Demo flow preparation.

Expected Impact & Scalability

IMPACT

- **Brand voice is locked in, not left to chance** — every output is RAG-grounded in the company's own documents, tone guidelines, and audience definitions. No drift, no off-brand surprises
- **Zero turnaround time eliminates the marketing bottleneck** — a feature ships and the complete content package (copy, visuals, video, email) is ready under 2 minutes. No waiting on designers, copywriters, or agencies. The launch moment is never wasted.
- **Personalized email campaigns reach every lead with unique subject lines and CTAs** — not one generic blast, but individually tailored messages at scale

SCALABILITY

- **Format expansion:** New content types (Reels, Stories, Shorts, Polls) are pluggable generator modules — drop into the pipeline without touching existing generators
- **Data growth:** Qdrant handles millions of brand memory vectors; PostgreSQL + Prisma handles relational growth (campaigns, leads, analytics) via zero-downtime migrations
- **Compounding intelligence:** Every published campaign feeds engagement data back into RAG memory — the system's output quality improves with every feature launch, turning usage into a moat.



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THANK YOU

Figma Based Design Prototypes.

UI Design Prototypes – Landing, Onboarding & Configuration

